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Terahertz metasurface sensor and its applications

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Terahertz metasurface sensor and its applications

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Abstract

Terahertz waves, often referred to as far-infrared radiation, represent a unique band of high-frequency electromagnetic waves. In biosensing applications, terahertz waves possess extremely low photon energies—much lower than those of visible and near-infrared radiation—thereby ensuring the safety of biological detection. At the same time, their frequency range covers the characteristic spectral regions associated with collective vibrational modes and weak intermolecular interactions of biological macromolecules. In addition, terahertz waves exhibit strong penetrability through dielectric materials, and a single terahertz pulse can span an ultrabroad bandwidth ranging from the gigahertz to tens of terahertz. When integrated with metasurfaces, terahertz metasurface sensors can further enhance local electric fields, suppress background noise and interference, significantly improve the detection sensitivity, and enable high-sensitivity quantitative analysis of trace biomolecules. This review provides a comprehensive overview of terahertz metasurface sensors from multiple perspectives. It covers terahertz refractive index sensors based on various operating principles, metasurfaces fabricated from different materials—including metallic structures, doped semiconductors, and all-dielectric components—as well as hybrid designs incorporating two-dimensional materials like graphene. The article further categorizes sensors based on their specific application targets. Furthermore, it examines the different device architectures and modulation principles employed, analyzing their respective advantages, drawbacks, and future development trajectories.

1. Introduction

Terahertz radiation refers to the electromagnetic waves within the frequency range of 0.1–10 THz (wavelength: 0.03–3 mm), situated in the transitional region between microwaves and infrared light [1–3]. Terahertz waves can penetrate non-polar materials such as textiles, plastics, and ceramics, while being strongly reflected by metals and absorbed by water [4, 5]. This property makes it highly suitable for security screening, enabling non-contact detection of concealed threats like knives and explosives beneath clothing. Moreover, terahertz radiation possesses photon energies that are approximately one millionth of those of x-rays, and therefore does not damage biological molecular structures, enabling truly non-destructive testing [6]. On the other hand, the vibrational and rotational energy levels of biological molecules, such as proteins and DNA, lie precisely within the terahertz frequency range. By analyzing characteristic absorption features, researchers can accurately identify the composition of substances and even distinguish active pharmaceutical ingredients from excipients, thereby providing powerful tools for medical diagnostics and drug quality control. In addition, the terahertz band offers a bandwidth more than three orders of magnitude wider than that of microwaves, supporting theoretical transmission rates exceeding 100 Gbps. This highlights the disruptive potential of terahertz technology in emerging applications such as 6 G communications and space exploration [7, 8].

However, terahertz waves interact only weakly with most natural materials, resulting in conventional devices that suffer from low efficiency, bulky configurations, and limited functionality, making them

difficult to meet the growing demands for miniaturization, integration, and multifunctionality in modern technologies. As two-dimensional artificial electromagnetic structures, metasurfaces enable flexible manipulation of the amplitude, phase, polarization, and other properties of terahertz waves through the precise design of subwavelength meta-atoms. This strategy overcomes the intrinsic limitations of natural materials and allows the realization of diverse functionalities, including efficient absorption, spectral filtering, polarization control, and holographic imaging. So, researchers have developed terahertz metasurface sensors that combine the advantages of terahertz radiation with the structural versatility of metasurfaces. These sensors offer high sensitivity, high resolution, and label-free detection capabilities. Through precise manipulation of electromagnetic waves at the subwavelength scale, they significantly enhance light–matter interactions, enabling sensitive detection of changes in the dielectric environment. This allows for efficient identification and analysis of biological samples, hazardous substances, and material properties [9]. Furthermore, metasurfaces can be integrated with various technologies such as microfluidics and electromagnetic modulation, greatly expanding their application scope. As a result, the combination of terahertz technology and metasurfaces demonstrates transformative potential across medical, environmental, and biological fields [10].

This article reviews the development of terahertz metasurface sensors, exploring diverse detection methods and refractive index sensing schemes based on various physical mechanisms. It systematically analyzes how material choice—from metals and doped nanoparticles to all-dielectric and 2D materials—affects sensor performance. The review also compares the adaptability of terahertz metasurface sensors to targets such as biomolecules, nucleic acids, pathogens, and pesticides. Finally, this article provides an outlook on the development prospects of terahertz metasurface sensors.

2. Principle of terahertz metasurface sensors

At present, most well-known terahertz metasurface sensors are refractive index-based devices. Its sensing mechanism enables accurate identification of material composition by monitoring resonance frequency shifts induced by variations in the analyte's refractive index. Refractive index unit (RIU) is a dimensionless unit used to quantify variations in refractive index. In refractive-index-based sensing applications, RIU is commonly employed to express sensor sensitivity, which is defined as the change in resonance frequency or wavelength per unit change in refractive index (THz RIU^{-1} or nm RIU^{-1}). Researchers have widely applied several resonance enhancement techniques to improve sensitivity, including inductance capacitance (LC) resonance, electromagnetic induction transparency (EIT), quasi bound states in continuum (QBIC), and Fano resonance.

Key advantages of terahertz refractive index sensors include their label-free operation, non-destructive nature, and strong anti-interference capability. They can be adapted to a variety of scenarios involving biological samples, chemical substances, and environmental gases. Integration with functional materials such as graphene and perovskites further enables dynamic control of chemical potential and wide-angle response. Currently, this technology has demonstrated promising potential in applications such as water quality monitoring, medical diagnosis, and safety inspection. Looking forward, through advanced electromagnetic field optimization strategies, further breakthroughs in sensitivity and resolution are anticipated.

2.1. Terahertz refractive index sensors based on LC resonance

LC resonance is the most prevalent resonance mechanism employed in terahertz metasurfaces. Refractive index sensors based on LC resonance constitute an important class of opto-electrical sensing devices. Compared with conventional optical refractive index sensors, LC-resonant sensors offer several inherent advantages, including strong immunity to electromagnetic interference, compact structural design, and high compatibility with on-chip integration. Their operating principle is based on monitoring resonance frequency shifts of the LC circuit, which indirectly reflect variations in the refractive index of the surrounding medium.

A typical LC resonant sensor consists of a planar inductive loop coupled with a capacitive gap structure. When the metasurface interacts with an analyte, changes in the refractive index modify the effective dielectric constant within the capacitive region, leading to a variation in capacitance. According to the LC resonance condition, even a slight capacitance change can induce a pronounced shift in the resonant frequency. Consequently, accurate measurement of the frequency shift enables precise determination of the refractive index of the target material.

In 1999, Pendry and co-workers at Imperial College London proposed the split-ring resonator (SRR), a microstructured unit cell composed of concentric metallic rings with narrow gaps, which exhibits a

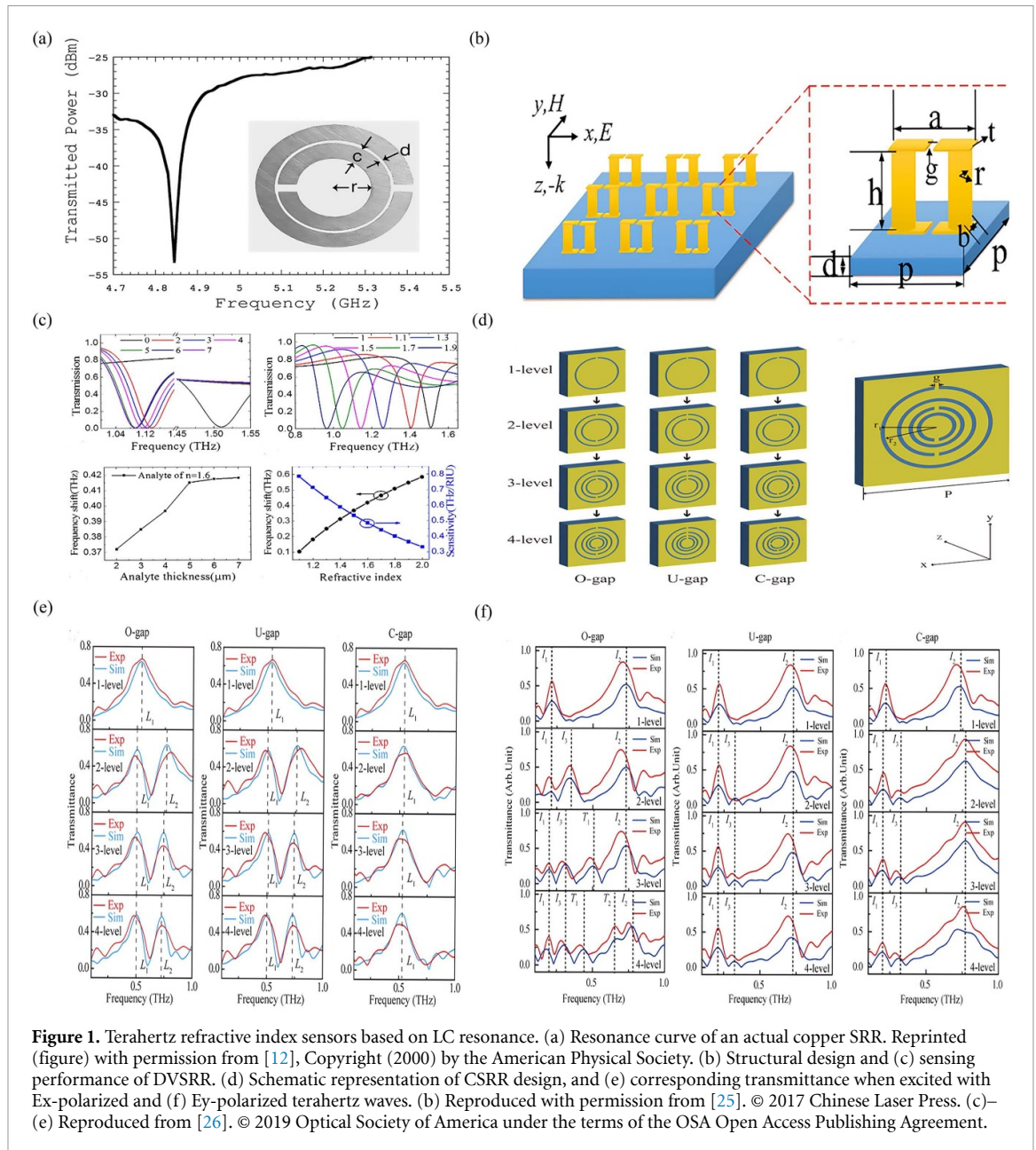


Figure 1. Terahertz refractive index sensors based on LC resonance. (a) Resonance curve of an actual copper SRR. Reprinted (figure) with permission from [12], Copyright (2000) by the American Physical Society. (b) Structural design and (c) sensing performance of DVSSRR. (d) Schematic representation of CSRR design, and (e) corresponding transmittance when excited with Ex-polarized and (f) Ey-polarized terahertz waves. (b) Reproduced with permission from [25]. © 2017 Chinese Laser Press. (c)–(e) Reproduced from [26]. © 2019 Optical Society of America under the terms of the OSA Open Access Publishing Agreement.

pronounced magnetic response [11]. The SRR represents a canonical structure for exciting LC resonance and has been widely adopted in metasurface-based sensing applications. As illustrated in figure 1(a), a typical SRR consists of a metallic square ring with a narrow split on one side, exhibiting a resonance peak at 4.845 GHz [12]. Sensors based on the SRR structure achieve a local field-enhanced resonance mode through the LC interaction [13–15]. In this metasurface design, the metallic portion functions as an inductor, while the gap acts as a capacitor [16, 17]. In this configuration, the metallic ring functions as an inductor, while the split gap acts as a capacitor. When the incident electromagnetic wave is polarized parallel to the gap, an LC resonance mode is excited [18–20]. At resonance, counter-propagating surface currents circulate along the ring, resulting in charge accumulation across the gap and generating a strongly localized electromagnetic field [21]. This intense field confinement near the gap region significantly enhances the interaction between the resonator and the surrounding dielectric environment, thereby enabling high-sensitivity refractive index sensing [22–24]. Despite its flexibility and design simplicity, the conventional SRR suffers from limitations such as narrow operating bandwidth, sensitivity to fabrication imperfections, and increased losses at higher frequencies, motivating continuous structural optimization.

To address these challenges, Wang *et al* proposed a high-sensitivity metasurface sensor based on a symmetric dual-slot vertically opened SRR (DVSSRR) in 2017, as shown in figures 1(b) and (c) [25]. The

introduction of structural symmetry effectively suppresses bi-isotropy, facilitating the excitation of high-quality-factor (high-Q) LC resonance modes. As a result, the Q-factor of the DVSRR is approximately 20 times higher than that of conventional LC resonators. Furthermore, elevating the resonator above the substrate substantially reduces dielectric losses introduced by the supporting material. Benefiting from these combined effects, the DVSRR-based sensor exhibits remarkable sensitivity to analyte-induced refractive index changes, achieving a maximum sensitivity of $0.788 \text{ THz RIU}^{-1}$.

Further advancements were reported by Gu and co-workers in 2019, who systematically investigated SRR-based metasurfaces with different symmetry-breaking configurations to overcome the sensitivity limitations of conventional and symmetric SRR designs. This structure, referred to as the complementary SRR (CSRR), is illustrated in figure 1(d) [26]. Three distinct gap arrangements were introduced: the O-gap configuration, in which adjacent SRRs are rotated by π radians; the U-gap configuration, featuring unidirectionally aligned gaps; and the C-gap configuration, where adjacent SRRs are rotated by $\pi/2$ radians clockwise. For terahertz waves polarized perpendicular to the x -axis, the transmission spectra shown in figure 1(e) reveal that both the U-gap and O-gap structures support two eigenmodes, whereas only a single eigenmode is observed in the C-gap configuration. The corresponding Q-factors increase with the number of ring layers. When the polarization is rotated perpendicular to the y -axis, as depicted in figure 1(f), three eigenmodes emerge in all three configurations once the ring number exceeds one. Notably, when the number of rings reaches four, two trapped modes appear in the O-gap structure. All eigenmodes and trapped modes exhibit slight redshifts in frequency as the structural complexity increases. Analysis of surface current and magnetic field distributions indicates that the dominant eigenmodes originate from localized LC resonances in the first- or second-order SRRs, while the trapped modes arise from hybrid coupling between higher-order localized dipole oscillations and multiple LC resonances. This symmetry-engineering strategy provides a promising route for the development of advanced metasurface-based biosensors and high-resolution thin-film sensing platforms.

2.2. Terahertz refractive index sensor based on EIT resonance

The EIT effect also arises from the multimodal coupling in terahertz metasurfaces. Specifically, it results from the interaction between bright and dark modes. Bright modes, such as dipole resonances in metallic rods, are resonant modes that can be directly excited by the incident field. In contrast, dark modes are generally not directly excited by the incident field but can be coupled via the alternating field generated by the bright modes. Since bright and dark modes typically originate from distinct structural elements, the unit cells of EIT metasurfaces often consist of composite rather than isolated shapes. If the structures supporting bright and dark modes are separated and subjected to the same external field, the dark-mode structure exhibits no electromagnetic response, as it can only be excited indirectly through the near-field coupling from the bright mode [27, 28]. Refractive index sensors based on EIT resonance are highly sensitive optical sensing devices. Their core operating principle relies on quantum interference effects that induce transparency in a specific frequency window. When two coherent light beams (pump and probe) simultaneously interact with the medium, destructive quantum interference suppresses the absorption of the probe light, resulting in a sharp transmission peak. The spectral position of this transparency peak is highly sensitive to changes in the surrounding refractive index. When a target analyte interacts with the sensor surface, the resulting refractive index variation induces a spectral shift of the transparency peak. By monitoring this shift, the refractive index of the analyte can be accurately determined.

Cai *et al* report a terahertz metamaterial that exploits the EIT effect [29], as depicted in figure 2(a). Through near-field coupling between a bright mode and a dark mode [30–32], a sharp transparency window emerges in the terahertz transmission spectrum [33–36]. However, a major drawback of most metasurface-based EIT sensors is their restriction to a single operational frequency band [37–39], which considerably narrows their potential in sensing applications [40–42]. To overcome this limitation, Zhu *et al* introduced an improved metasurface design incorporating a metal rod, two fork-shaped structures (FSS), and a SRR [43]. As shown in figure 2(b), numerical simulations confirm that this structure supports dual-band EIT resonances at 0.89 THz and 1.56 THz. A notable feature of this sensor is that each resonance exhibits distinct sensing characteristics tailored to analytes with different refractive indices. Specifically, the first resonance is suitable for detecting analytes with refractive indices ranging from 1.5 to 2, such as the MS2 virus ($n \approx 1.97$), PRD1 virus ($n \approx 1.87$) [44], and polymethyl methacrylate ($n \approx 1.6$) [45]. In contrast, the second resonance is optimized for sensing analytes with refractive indices between 1.0 and 1.5, including various molds ($n \approx 1.11$ – 1.36) [46].

Furthermore, the EIT effect can be realized not only through bright–dark mode coupling but also via the interaction between two identical types of modes. As shown in figure 2(c) [47], Wu *et al* propose a terahertz asymmetrically coupled resonator (ACR) consisting of two different SRRs was designed. A

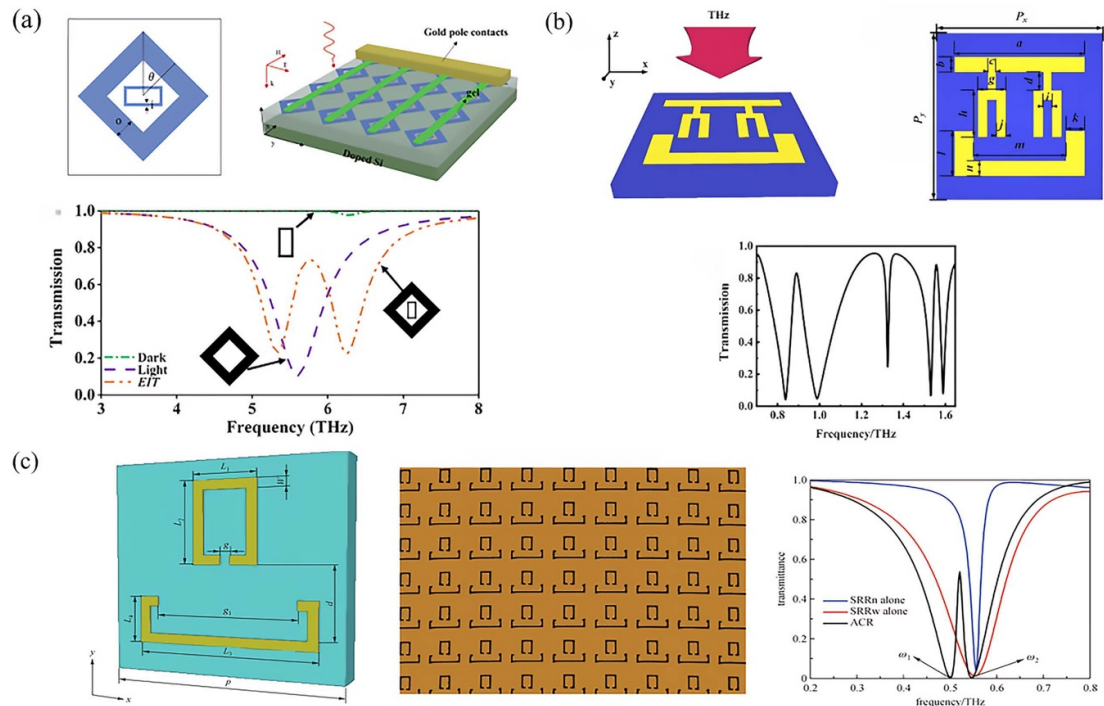


Figure 2. Terahertz refractive index sensor based on EIT resonance. (a) Structural design of metasurface and transmission curve of each part of the device. Reprinted from [29], Copyright (2020), with permission from Elsevier. (b) A metasurface composed of one metal rod, two FSS, and one SRR. Reproduced from [43]. © 2021 Optical Society of America under the terms of the OSA Open Access Publishing Agreement. (c) Resonance curve of ACR metasurface. Reproduced from [47], with permission from Springer Nature.

narrow SRR and a wide SRR exhibit transmission peaks at similar spectral positions, yet they differ significantly in their quality (Q) factors. When these two modes are coupled, sharper transmission peaks arise as a consequence of resonance hybridization. In this configuration, the asymmetric resonant coupling results from the interaction between two excited plasmonic modes. Both the resonance intensity and Q -factor are affected by the inter-structure spacing. Sensors leveraging such EIT-based coupling mechanisms generally feature sharper resonance peaks and higher Q -factors, thus supporting enhanced sensing performance.

2.3. Terahertz refractive index sensor based on quasi-bound state in the continuum (QBIC) resonance

The BIC effect occurs when the radiation field of a metasurface remains entirely confined within the structure without leakage, rendering it undetectable in the far field. Under ideal conditions, a BIC possesses an infinite Q -factor and cannot be observed even with enhanced spectral resolution. To make such states measurable, structural perturbations are introduced—for instance, by modifying the parameters of a unit cell with period p and expanding the metasurface period from 1 to p . This breaks the symmetry and converts the ideally bound state into a radiating one, leading to far-field emission and a sharp dip in the transmission spectrum. This observable phenomenon is referred to as a QBIC. The refractive index sensor based on QBIC resonance is an emerging device that leverages metasurface structures to achieve highly sensitive optical detection. Its operating principle relies on exciting QBIC resonances with ultra-high Q -factors through carefully designed asymmetry in the metasurface unit, which induces interference coupling between electric quadrupole and magnetic dipole modes. This resonant mode exhibits extreme sensitivity to changes in the surrounding medium's refractive index. Any variation in the refractive index induces a significant shift in the resonance wavelength, allowing the analyte's refractive index to be determined by tracking this spectral shift.

The metasurface shown in figure 3(a) is composed of a periodic array of aluminum double split rectangular resonators (DSRRs) with a double-elliptical geometry, fabricated on a silicon dioxide substrate [48]. Structural symmetry breaking is deliberately introduced by tuning the geometric parameter g_2 , which enables effective modulation of the resonant response. To facilitate practical liquid-phase detection, the metasurface sensor is further integrated with microfluidic channels, allowing controlled interaction between the resonant structures and the target analyte. Symmetry breaking

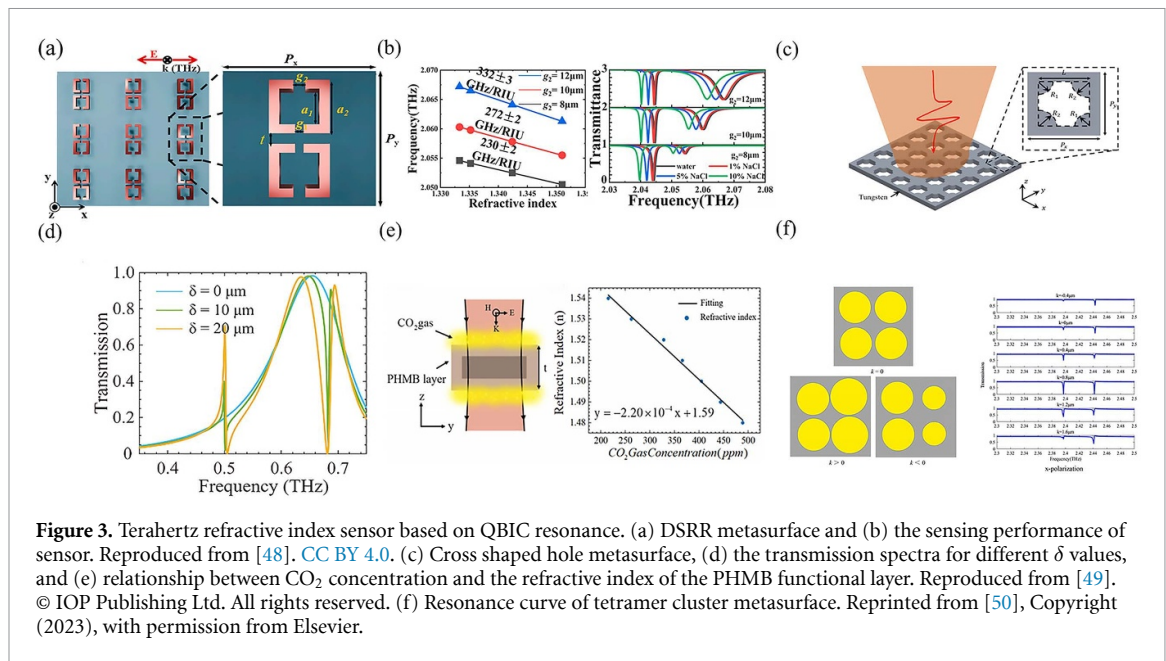


Figure 3. Terahertz refractive index sensor based on QBIC resonance. (a) DSRR metasurface and (b) the sensing performance of sensor. Reproduced from [48], CC BY 4.0. (c) Cross shaped hole metasurface, (d) the transmission spectra for different δ values, and (e) relationship between CO_2 concentration and the refractive index of the PHMB functional layer. Reproduced from [49]. © IOP Publishing Ltd. All rights reserved. (f) Resonance curve of tetramer cluster metasurface. Reprinted from [50], Copyright (2023), with permission from Elsevier.

is achieved by varying the parameter g_2 , and the sensor is integrated with microfluidic channels for liquid sensing. As illustrated in figure 3(b), an increase in the liquid refractive index from 1.333 to 1.351 induces a redshift in the resonance frequency. The maximum absorption rates of the DSRR in QBIC mode are 47%, 44%, and 19% for g_2 values of 12, 10, and 8 μm , respectively. The corresponding resonance frequencies lie between 2.06–2.07 THz ($g_2 = 12 \mu\text{m}$), 2.054–2.064 THz ($g_2 = 10 \mu\text{m}$), and 2.047–2.057 THz ($g_2 = 8 \mu\text{m}$). In terms of sensitivity, the average refractive index sensitivity S is 0.332 ± 0.003 , 0.272 ± 0.002 , and $0.230 \pm 0.002 \text{ THz RIU}^{-1}$ for $g_2 = 12$, 10, and 8 μm , respectively. Among these, the configuration with $g_2 = 12 \mu\text{m}$ exhibits superior linearity in refractive index sensitivity compared to the others. This enhanced linearity ensures more accurate and consistent detection, underscoring the suitability of the metasurface sensor for liquid sensing applications.

The refractive index sensor based on QBIC resonance is applicable not only for liquid sensing but also for gas detection. As illustrated in figure 3(c), Liu *et al* designed a novel polarization-insensitive terahertz metallic metasurface with cross-shaped openings [49]. Each cross-shaped aperture is formed by four fan-shaped notches with diagonal radii R_1 and R_2 . The structural asymmetry parameter δ ($\delta = R_2 - R_1$) is controlled by adjusting R_2 . When the mirror symmetry of the metasurface is broken, QBIC modes are excited, as shown in figure 3(d). For gas sensing applications, a CO_2 detection mechanism has been demonstrated using a metasurface functionalized with a poly (hexamethylene biguanide) (PHMB) polymer layer [51–53]. PHMB—a polymer rich in amide groups—serves as a selective host material for CO_2 adsorption, as its interactions with hydrogen, nitrogen, and other potential interferents are negligible, thereby ensuring high selectivity for CO_2 . Figure 3(e) illustrates the linear relationship between the refractive index n of the PHMB layer and the CO_2 concentration, attributed to the redistribution of electron density within the polymer layer upon CO_2 exposure. This finding has facilitated the development of high-performance terahertz metasurfaces for environmental gas monitoring, nonlinear optics, and integrated photonic systems.

To further enhance the quality factor, Wang *et al* proposed an all-dielectric metasurface composed of tetramer clusters supporting dual terahertz resonances [50]. In this design, quartz serves as the substrate, while silicon is employed to construct the tetramer clusters. By selectively tuning the radii of the cylindrical elements within the metasurface lattice, structural asymmetry is introduced, enabling the excitation of QBICs. This is a relatively novel process to employ for symmetry breaking. The asymmetric parameter k represents the variation of the radius in both cylinders. When the asymmetry parameter $k = 0$, all four cylinders retain identical dimensions. When $k > 0$, the radii of two cylinders increase, whereas when $k < 0$, the radii of two cylinders decrease. Notably, variations in k do not alter the center coordinates of any cylinder. When deployed as a refractive index sensor, the metasurface exhibits optimal performance at an asymmetry parameter of $k = 0.8 \mu\text{m}$, achieving a Q -factor of 2303.77. This strategy of tuning the radii within the tetramer cluster to break structural symmetry and enhance the Q -factor offers a promising direction for advancing terahertz sensing technologies.

2.4. Terahertz refractive index sensor based on Fano resonance

The principle of Fano resonance can be briefly described as resulting from the breaking of symmetry within a unit structure [54]. Specifically, when a metasurface unit cell exhibits axisymmetric distribution with respect to the polarization state of the incident wave, the corresponding transmission spectrum displays a broad, deep, and approximately symmetric U-shaped curve, known as a Lorentzian lineshape. Under such conditions, the spatial scale of photon distribution is smaller than the radiation wavelength, and atomic radiation undergoes coherent superposition, leading to an increased radiation rate—a state referred to as superradiance [55, 56]. Conversely, when atomic radiation experiences coherent cancellation, the radiation rate decreases significantly, resulting in a subradiant state. Such states are commonly observed in higher-order modes of plasmonic resonances. Fano resonance arises precisely from the coupling between these two radiative states. When the symmetry of the unit cell is broken, the condition for radiation coupling is satisfied, and a sharp V-shaped absorption dip emerges alongside the U-shaped absorption profile. At this point, the symmetry of the spectral lineshape is also broken, giving rise to the characteristic Fano lineshape. Refractive index sensors based on Fano resonance are high-sensitivity optical sensing devices that leverage the coupling between discrete and continuum states to generate asymmetric spectral lines [25, 57, 58].

For the description of Fano resonance, we still choose the example based on SRR technology because the SRR structure has flexible and adjustable geometric parameters, which can be precisely designed to excite a strong Fano resonance effect. Its unique open resonance characteristics lead to significant and sharp resonance peaks when terahertz waves interact with matter, greatly enhancing the sensitivity of the sensor to refractive index changes. At the same time, this structure has a mature research foundation in the terahertz band, facilitating experimental implementation and performance optimization. Fatima Taleb *et al* introduced an asymmetric SRR design that induces pronounced Fano resonance [59], as illustrated in figure 4(a). This planar metasurface, referred to as FCASRR, consists of four aluminum resonators whose symmetry is broken by designing two of the air arms to be longer than the others. The upper and lower arm lengths are defined as L_b and L_t , respectively. An asymmetry parameter is introduced as $\delta = [(L_t - L_b)/(L_t + L_b)] \times 100\%$. When the metasurface was applied to sense lactose and sucrose, the resulting sensing performance is shown in figure 4(b). The resonance frequencies shift by 73 GHz and 80 GHz for lactose and sucrose analytes, respectively. With increasing refractive index, a linear redshift in the Fano resonance is observed. The FCASRR structure exhibits a sensing sensitivity several times higher than that of the symmetric SRR configuration.

Building on this concept, Xie *et al* further modified a planar metasurface composed of four simple metal resonators to achieve a higher Q-factor [60]. As shown in figure 4(c), structural asymmetry was introduced by varying the parameter d . Figure 4(d) presents the transmission spectrum of the proposed metasurface, which exhibits two transmission dips at 0.430 THz and 0.809 THz, with transmittance values of 0.10% and 26.45%, respectively. These two resonant modes are denoted as Rs and Rd, where Rs corresponds to the resonance at 0.430 THz and Rd to the higher-frequency resonance. The Rs mode shows a symmetric Lorentzian lineshape with a relatively broad bandwidth of 0.256 THz. In contrast, the Rd mode displays a distinctly asymmetric Fano lineshape, which is significantly sharper, having a bandwidth of only 0.014 THz. The emergence of the asymmetric Fano profile is attributed to the interaction between dark and bright modes—specifically, the coupling of non-radiative states derived from radiative states with the continuum spectrum. The resulting Q-factor of this device reaches 58.

In addition to the basic SRR configuration, Pooja *et al* have developed an all dielectric metasurface refractive index sensor driven by dual Fano resonances [61]. The metasurface structure, as shown in figure 4(e), consists of three silicon rods with a thickness of t_2 and a width of w placed on a quartz substrate with a thickness of t_1 . Two additional silicon rods of equal length L_2 , located equidistant from the center silicon rod of length L_1 at the gap g , with a period $P = P_x = P_y$. The corresponding resonance modes are shown in figure 4(f), where the Q-factors for $f_1 = 1.79$ THz and $f_2 = 3.90$ THz are 920.00 and 685.53, respectively. This indicates that the design has excellent performance and is expected to become a promising tool in the field of sensing.

2.5. Comparison of terahertz metasurface sensors with different resonance mechanisms

Terahertz refractive index sensors based on different resonance mechanisms exhibit diverse performance characteristics, as summarized in table 1. LC resonant sensors operate through inductance–capacitance resonances in metallic structures, featuring relatively simple designs and good compatibility with integrated fabrication processes. However, their performance is generally limited by intrinsic metal losses, resulting in comparatively low Q-factors, which makes them more suitable for applications with relaxed precision requirements. EIT-based sensors generate narrow transmission windows via interference

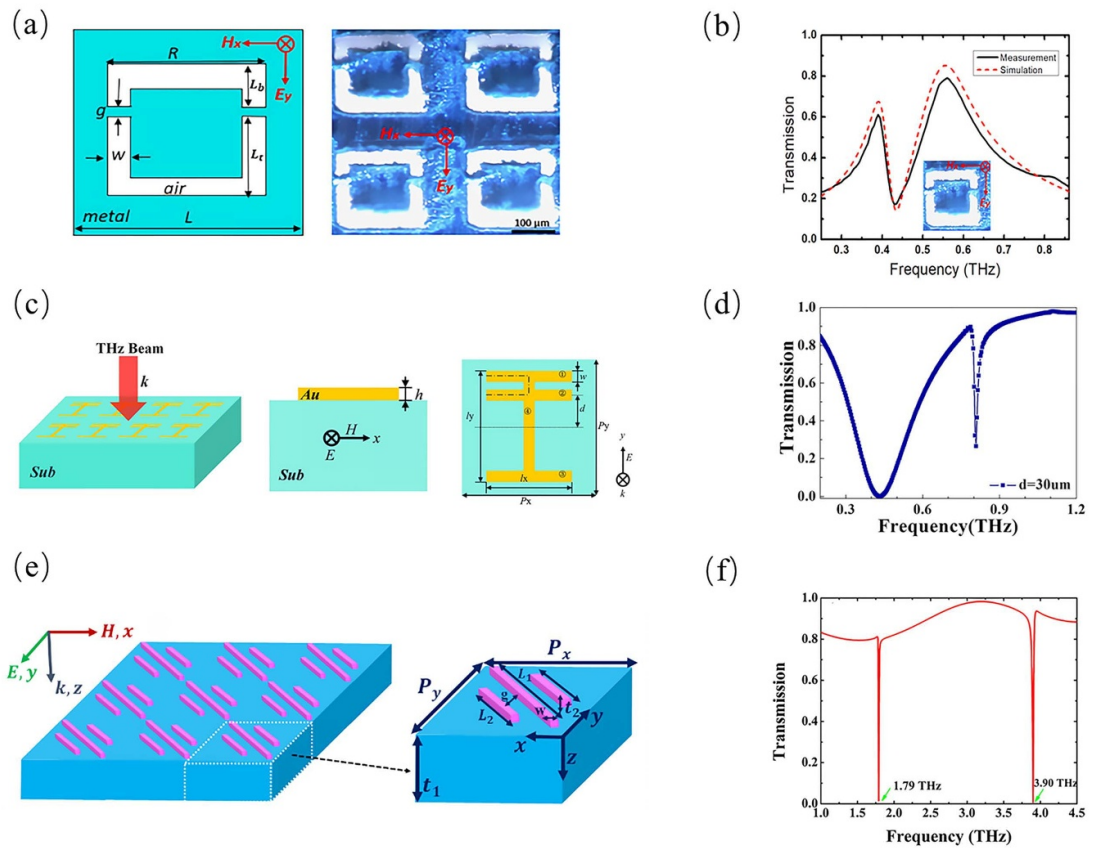


Figure 4. Terahertz refractive index sensor based on Fano resonance. (a) FCASRR metasurface and (b) its resonance curve. Reproduced from [59], CC BY 4.0. (c) Four-strip metallic metasurface and (d) corresponding resonance curve. Reproduced from [60], CC BY 4.0. (e) All dielectric metasurface and (f) corresponding resonance curve. © (2024) IEEE. Reprinted, with permission, from [61].

Table 1. Comparison of terahertz metasurface sensors with different resonance mechanisms.

	LC resonance [25]	EIT resonance [43]	QBIC resonance [50]	Fano resonance [59]
Operating frequency band	0.6–1.6 THz	0.8–1.7 THz	2.3–2.5 THz	0.3–0.8 THz
Sensitivity	0.788 THz RIU ⁻¹	0.201 THz RIU ⁻¹	0.067 THz RIU ⁻¹	0.092 THz RIU ⁻¹
Q-Factor	20	56.90	2303.77	29

between bright and dark modes, enabling enhanced sensing performance. Owing to their strong dependence on near-field coupling and structural precision, these sensors are typically employed in high-accuracy sensing scenarios, such as gas detection. QBIC sensors introduce structural asymmetry to access high-Q resonant states, and the associated strong local field enhancement provides favorable conditions for achieving high sensitivity, particularly in trace-level and biomolecular sensing. Fano resonant sensors are characterized by sharp asymmetric spectral responses arising from destructive interference between bright and dark modes. Their sensing performance can be effectively tuned through structural parameter optimization, offering greater flexibility for applications involving dynamic monitoring or multi-parameter sensing. In general, QBIC-based sensors demonstrate superior performance of Q-factor, whereas EIT and Fano resonant sensors present distinct advantages for specific sensing requirements.

3. Types of terahertz metasurface sensors

3.1. Terahertz metal metasurface sensor

Terahertz metal metasurface sensors typically consist of a dielectric substrate and a patterned metallic antenna array. Common substrate materials include quartz and silicon, while the antenna elements

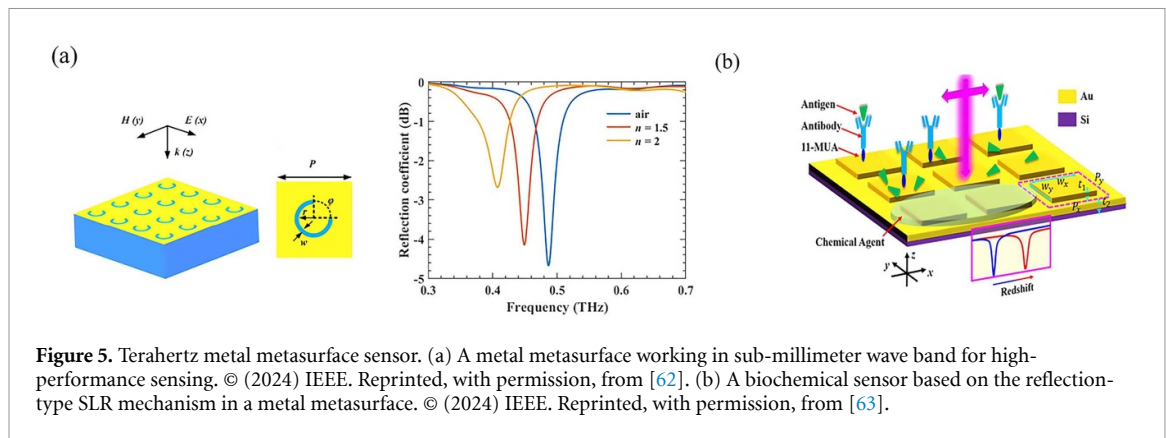


Figure 5. Terahertz metal metasurface sensor. (a) A metal metasurface working in sub-millimeter wave band for high-performance sensing. © (2024) IEEE. Reprinted, with permission, from [62]. (b) A biochemical sensor based on the reflection-type SLR mechanism in a metal metasurface. © (2024) IEEE. Reprinted, with permission, from [63].

are usually fabricated from highly conductive metals such as gold or copper. By carefully tailoring the geometry, size, and periodic arrangement of the metallic resonators, the electromagnetic response of the metasurface can be precisely engineered, enabling controllable resonance characteristics in the terahertz regime. These resonances can be effectively described using equivalent circuit models composed of inductive, capacitive, and resistive elements, providing a clear physical framework for sensing applications.

Within this context, He and Shao proposed a periodic metal metasurface, as shown in figure 5(a), in which the engineered surface patterns give rise to resonance behavior that can be modeled as a parallel RLC circuit [62]. Variations in the surrounding dielectric environment induce measurable shifts in the resonant frequency, allowing the metasurface to function as a high-performance refractive index sensor. This work demonstrates the fundamental sensing principle of metal metasurfaces based on resonance modulation.

Building upon the same concept of resonance engineering, Li *et al* further extended metal metasurface sensing to biochemical detection by exploiting surface lattice resonances (SLRs), as illustrated in figure 5(b) [63]. Operating in the near-infrared band, the reflective metal metasurface supports collective lattice resonances with reduced radiative loss, resulting in significantly enhanced sensing performance compared to conventional plasmonic sensors. The proposed sensor achieves a high sensitivity of $796.6 \text{ nm RIU}^{-1}$, with detection limits of 0.5% and 1% for sodium chloride and glucose solutions, respectively. Moreover, through surface functionalization, the metasurface enables specific and efficient detection of carcinoembryonic antigen at low concentrations of 0.5 g ml^{-1} , highlighting its potential for practical biochemical sensing applications.

3.2. Terahertz metasurface sensor doped with metal nanoparticles

Building upon conventional metal metasurface sensors, terahertz metasurface sensors incorporating metal nanoparticles still rely on the interaction between metallic structures and electromagnetic waves, yet introduce additional degrees of freedom for field manipulation. Unlike planar metal metasurfaces that primarily exploit geometrically engineered resonances, the integration of metal nanoparticles leverages localized surface plasmon resonance (LSPR) to further enhance electromagnetic field confinement at the sensing interface. This synergistic coupling between metasurface resonances and nanoparticle-induced LSPR significantly improves sensing sensitivity and specificity.

By introducing metal nanoparticles, the performance of traditional metal metasurfaces can be optimized and functionally extended, thereby broadening their application scope to areas such as biosensing and environmental monitoring. Metal nanoparticles provide substantial advantages for terahertz metasurface sensors [64, 65]. Specifically, their pronounced LSPR effect strongly enhances the local electromagnetic field, enabling the detection of weak signals that are otherwise difficult to resolve [66]. In addition, nanoparticles with diverse sizes, shapes, and materials can support multi-frequency responses, effectively expanding the operational bandwidth and detection range of the sensor. Through appropriate surface functionalization, metal nanoparticles can be selectively engineered to bind with multiple target analytes, improving molecular recognition capability. Furthermore, their tunable optical properties contribute to background noise suppression and enhanced signal contrast, which strengthens the sensor's anti-interference performance in complex environments and leads to overall performance optimization [67–69].

Conventional terahertz detection techniques face inherent challenges in detecting trace-level samples, primarily due to the weak interaction between terahertz waves and matter. Consequently, strategies that

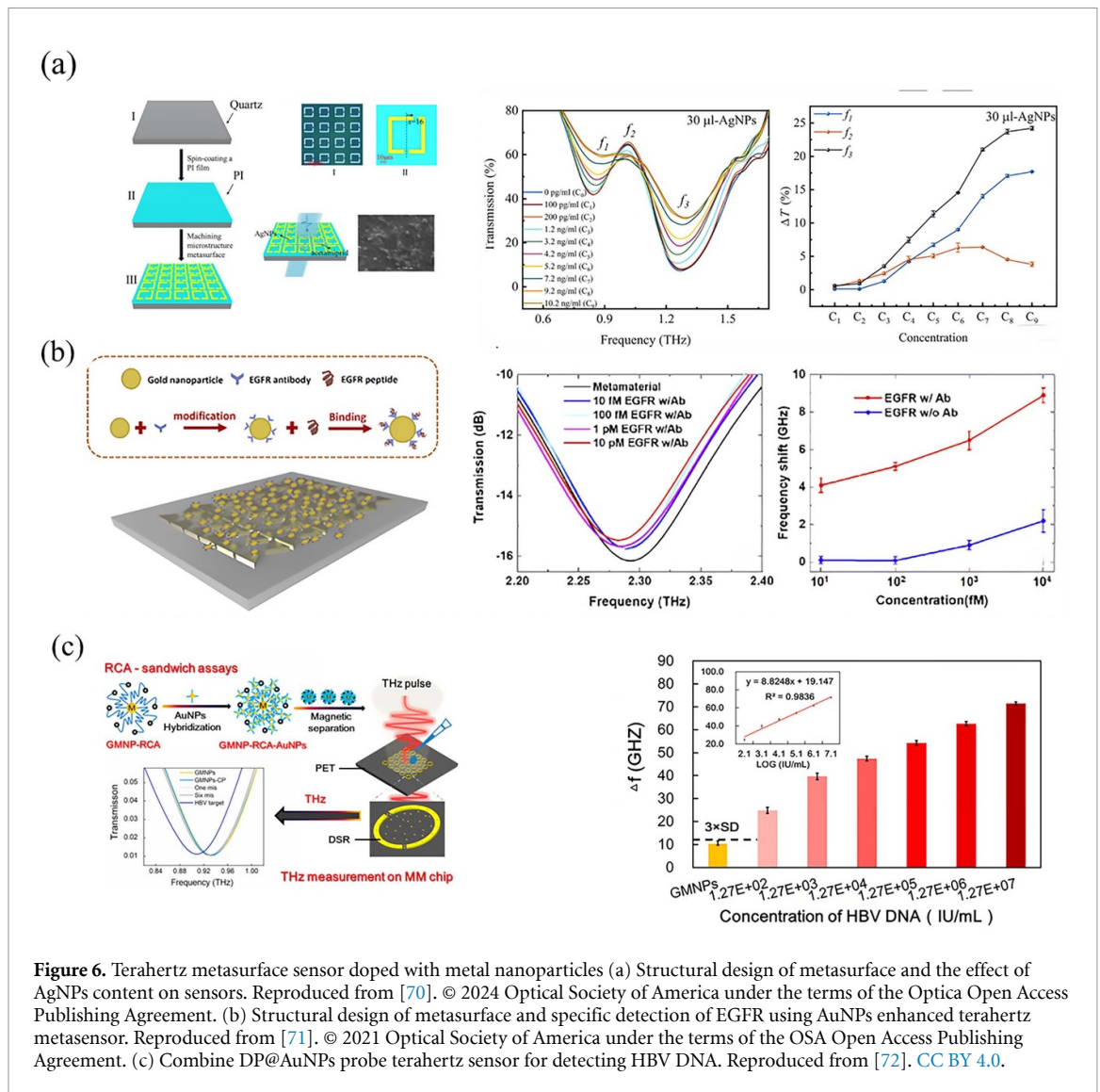


Figure 6. Terahertz metasurface sensor doped with metal nanoparticles (a) Structural design of metasurface and the effect of AgNPs content on sensors. Reproduced from [70]. © 2024 Optical Society of America under the terms of the Optica Open Access Publishing Agreement. (b) Structural design of metasurface and specific detection of EGFR using AuNPs enhanced terahertz metasensor. Reproduced from [71]. © 2021 Optical Society of America under the terms of the OSA Open Access Publishing Agreement. (c) Combine DP@AuNPs probe terahertz sensor for detecting HBV DNA. Reproduced from [72]. CC BY 4.0.

locally enhance terahertz electromagnetic fields have become essential for improving sensing performance. One effective approach is the integration of metal nanoparticles with metasurfaces, enabling synergistic field enhancement through localized plasmonic effects.

Yan *et al* proposed and experimentally demonstrated a Fano-resonance metasurface sensor integrated with silver nanoparticles (FaMs-AgNPs), as illustrated in figure 6(a) [70]. The incorporation of AgNPs leads to charge accumulation and pronounced local electric field enhancement via the tip effect, thereby strengthening the interaction between terahertz waves and analytes and improving sensor sensitivity. The authors systematically investigated the influence of AgNP concentration on the detection of imidacloprid, considering four different volumes: 10, 20, 30, and 40 μl . An optimal concentration of 30 μl yielded the largest amplitude modulation and highest sensitivity, enabling detection of imidacloprid at concentrations as low as 100 pg m^{-1} . Owing to its simple structure, ease of fabrication, and excellent sensing performance, the FaMs-AgNPs sensor demonstrates strong potential for terahertz trace detection applications.

While silver nanoparticles offer effective electromagnetic field enhancement, gold nanoparticles (GNPs) have been increasingly adopted in terahertz metasurface sensors due to their superior chemical stability, biocompatibility, and well-controlled surface plasmon resonance characteristics. Building upon the same nanoparticle-assisted enhancement principle, Li *et al* developed a bow-tie array terahertz metasurface biosensor functionalized with GNPs and epidermal growth factor receptor (EGFR) antibodies for specific biomolecular detection, as shown in figure 6(b) [71]. The authors systematically examined the effect of GNP size on sensitivity enhancement, revealing that metasurfaces modified with larger-diameter

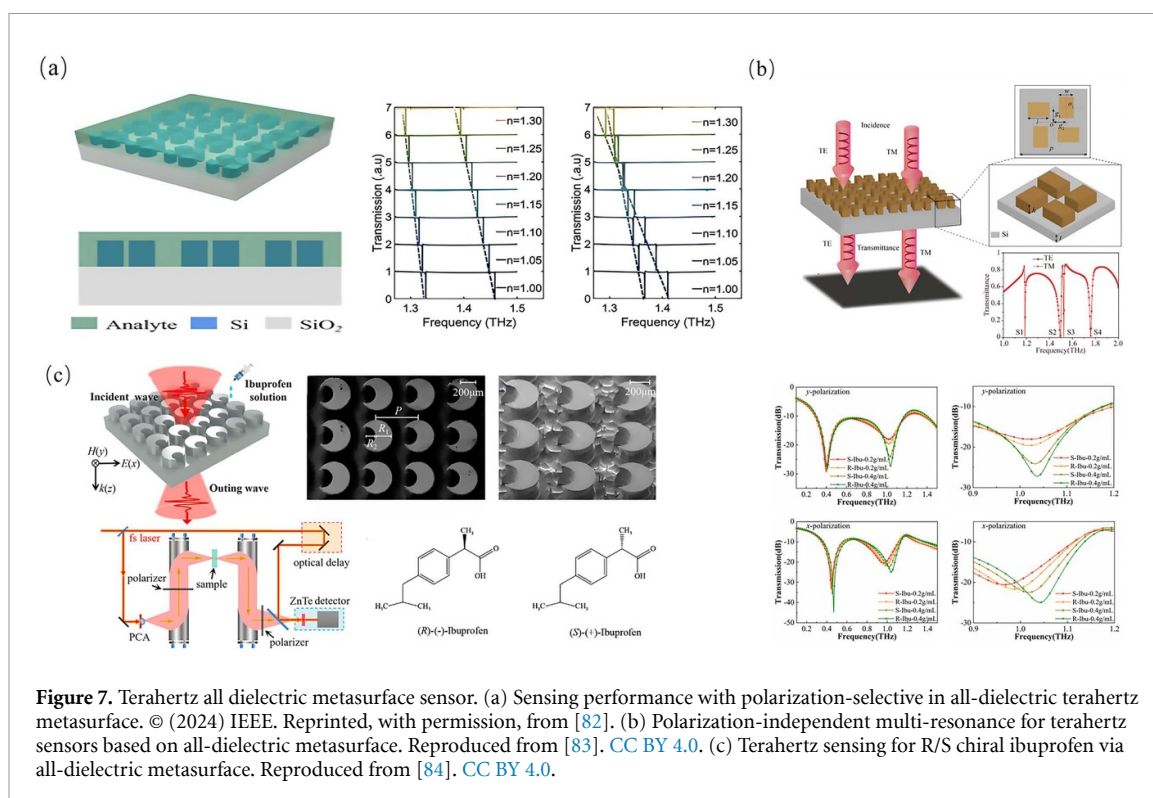


Figure 7. Terahertz all dielectric metasurface sensor. (a) Sensing performance with polarization-selective in all-dielectric terahertz metasurface. © (2024) IEEE. Reprinted, with permission, from [82]. (b) Polarization-independent multi-resonance for terahertz sensors based on all-dielectric metasurface. Reproduced from [83]. CC BY 4.0. (c) Terahertz sensing for R/S chiral ibuprofen via all-dielectric metasurface. Reproduced from [84]. CC BY 4.0.

GNPs produce greater resonance frequency shifts. The synergistic combination of GNPs and EGFR antibodies significantly improves EGFR sensing performance, highlighting the advantages of AuNP-based platforms for biosensing applications.

Extending this strategy further, Li *et al* introduced a nucleic-acid-based amplification scheme by immobilizing long single-stranded DNA (ssDNA) onto GNPs to form detection probes (DP@AuNPs), which were subsequently assembled into a GMNPs-RCA-AuNPs sandwich complex [72]. When integrated into a flexible terahertz metasensor chip, as depicted in figure 6(c), the complex induced a clear red shift in the metasurface resonance peak, enabling quantitative detection of hepatitis B virus (HBV) DNA. The biosensor exhibited excellent specificity and sensitivity, successfully discriminating single-base mismatches in HBV DNA within clinical serum samples. A low detection limit of 1.27×10^2 IU ml⁻¹ (international units per milliliter, a standard clinical metric for viral load quantification) was achieved, with a linear relationship observed between HBV DNA concentration and the resonance frequency shift over the range of 1.27×10^2 – 1.27×10^7 IU ml⁻¹.

3.3. Terahertz all dielectric metasurface sensor

All-dielectric metasurfaces, characterized by low loss, high Q-factor, and exceptional electromagnetic resonance control, have emerged as an ideal platform for enhancing the performance of terahertz sensors [73–75]. Through the design of subwavelength microstructures, they enable precise manipulation of the amplitude, phase, and polarization of electromagnetic waves, significantly strengthening light–matter interactions—particularly beneficial for high-sensitivity sensing [76, 77]. In contrast to metallic metasurfaces, all-dielectric counterparts employ low optical and thermal loss materials such as silicon [78], which effectively suppress both Ohmic and radiative losses. These properties support label-free, non-invasive detection and allow real-time monitoring of dynamic biological processes—such as molecular interactions and cellular metabolism—via appropriate functionalization. Moreover, the biocompatibility and excellent chemical stability of the constituent materials further broaden the potential applications of terahertz sensors across medical, environmental, and biological fields [79–81].

Conventional metallic metasurfaces suffer from inherent Ohmic losses, which restrict further improvement of their Q-factor. To address the limitations in terahertz refractive index sensing, Chen *et al* introduced a low-loss all-dielectric metasurface based on QBIC [82], as illustrated in figure 7(a). This structure supports multiple high Q resonance peaks. Owing to the unique field enhancement distribution enabled by the sensor design, the resonance peak exhibits a pronounced redshift as the refractive index of the analyte increases. Leveraging this behavior, the metasurface achieves a refractive index sensitivity of up to 0.442 THz RIU⁻¹ within the frequency range of 1.3–1.5 THz.

Based on a design principle of local asymmetry within global symmetry, all-dielectric metasurfaces can be extended to microwave or infrared bands. Zhang *et al* suggested that such designs could advance the application of terahertz metasurfaces in biosensing and medical detection. To this end, they proposed an all-dielectric metasurface capable of generating polarization-independent multiple high Q resonances [83], as depicted in figure 7(b). The metasurface consists of a windmill-like unit cell formed by four identical rectangular prisms. Each pair of adjacent prisms creates a locally asymmetric configuration that excites Fano resonance, while the overall central symmetry ensures polarization-independent response. The structure exhibits four polarization-independent resonances with high Q -factors, among which resonance S2 (1.4946 THz) achieves a frequency sensitivity of $0.273 \text{ THz RIU}^{-1}$, and resonance S4 (1.7587 THz) reaches $0.105 \text{ THz RIU}^{-1}$. Moreover, the transmission-structure and polarization independence of these resonances help to simplify experimental measurements.

All-dielectric metasurfaces have also demonstrated significant potential in chiral drug sensing. For instance, Shi *et al* developed a crescent-shaped cylindrical array terahertz all-dielectric metasurface for this purpose [84], as illustrated in figure 7(c). By monitoring resonance frequency shifts and transmittance changes, the metasurface sensor was used to detect (R)-(-)-ibuprofen and (S)-(+)-ibuprofen solutions coated on its surface. It achieved a maximum sensitivity of $0.060 \text{ THz mg}^{-1}$ for monitoring ibuprofen concentration. Furthermore, the sensor successfully discriminated between different chiral molecules—(R)-(-)-ibuprofen and (S)-(+)-ibuprofen—in trace samples as small as $5 \mu\text{l}$, with a maximum frequency difference of $0.018 \text{ THz mg}^{-1}$. These results validate the applicability of the crescent-shaped all-dielectric metasurface for enhanced sensing and detection of chiral molecules, offering a promising strategy for trace biomolecular recognition.

3.4. Metasurface sensors incorporating graphene or molybdenum disulfide

Terahertz metasurface sensors integrating graphene or molybdenum disulfide (MoS_2) exploit the distinctive electrical and optical properties of these materials, combined with the versatile electromagnetic control offered by metasurfaces [85–87]. Graphene and MoS_2 exhibit tunable conductivity and low losses in the terahertz frequency range [88, 89]. By dynamically modulating their carrier concentration through electrical, optical, or chemical doping, the resonant response of the metasurface can be precisely controlled [90]. Incorporating these materials into metasurfaces significantly enhances the local electromagnetic field [91], boosts sensitivity for trace analyte detection [92], and enables non-invasive, real-time monitoring [93]. These features make such hybrid systems highly promising for applications in biomedical imaging, environmental monitoring, and security screening [94–96].

Graphene has attracted considerable interest in terahertz biochemical sensing owing to its excellent biocompatibility, tunable properties, and remarkable optoelectronic characteristics. For example, Zhou *et al* proposed a novel multifunctional terahertz microfluidic platform based on a graphene-integrated complementary asymmetric split ring (CASR) metasurface as an efficient biosensing system for liquid environments [97], as shown in figure 8(a). This platform enables real-time, label-free molecular detection with high sensitivity and selectivity. Compared to conventional terahertz sensing approaches, this method eliminates the need for sample drying, offering a more convenient and direct detection process. By integrating graphene with the CASR metasurface within a microfluidic chip, the detection sensitivity is significantly enhanced. Furthermore, detecting analytes at the picogram level remains challenging in terahertz sensing. To address this, Chen *et al* developed a graphene-enhanced terahertz metasurface for label-free sensing [98]. As illustrated in figure 8(b), their sensor achieved the first quantitative detection of trace 1,3-DNB (which exhibits absorption at 2.52 THz), reaching a maximum reflection sensitivity of $10\% \text{ pmol}^{-1}$ and a detection limit of 42 pg. This sensing strategy extends the applicability of label-free terahertz sensing and strengthens its potential in pharmaceutical, environmental monitoring, and security applications.

MoS_2 , a representative two-dimensional transition metal dichalcogenide, offers multiple performance enhancements for terahertz sensors due to its layered structure and tunable bandgap characteristics. For instance, Tomas Pires *et al* developed a sensitive terahertz metasurface exhibiting EIT-like effects by incorporating a 2D MoS_2 layer, as illustrated in figure 8(c) [99]. At the resonant frequency of 2.02 THz, the metasurface achieves a Q -factor of 8.45—nearly double the value of 4.8 obtained with a conventional metasurface—along with a theoretical sensitivity of $0.661 \text{ THz RIU}^{-1}$. These results confirm that the introduction of the 2D MoS_2 layer significantly improves both the Q -factor and sensitivity of the terahertz metasurface, establishing it as a promising platform for applications such as protein biosensing.

3.5. Terahertz metasurface sensor combined with carbon nanotubes (CNTs)

CNTs offer significant advantages for terahertz metasurface sensors, owing to their high surface area, excellent biocompatibility, tunable properties, and unique photoelectric response in the terahertz band

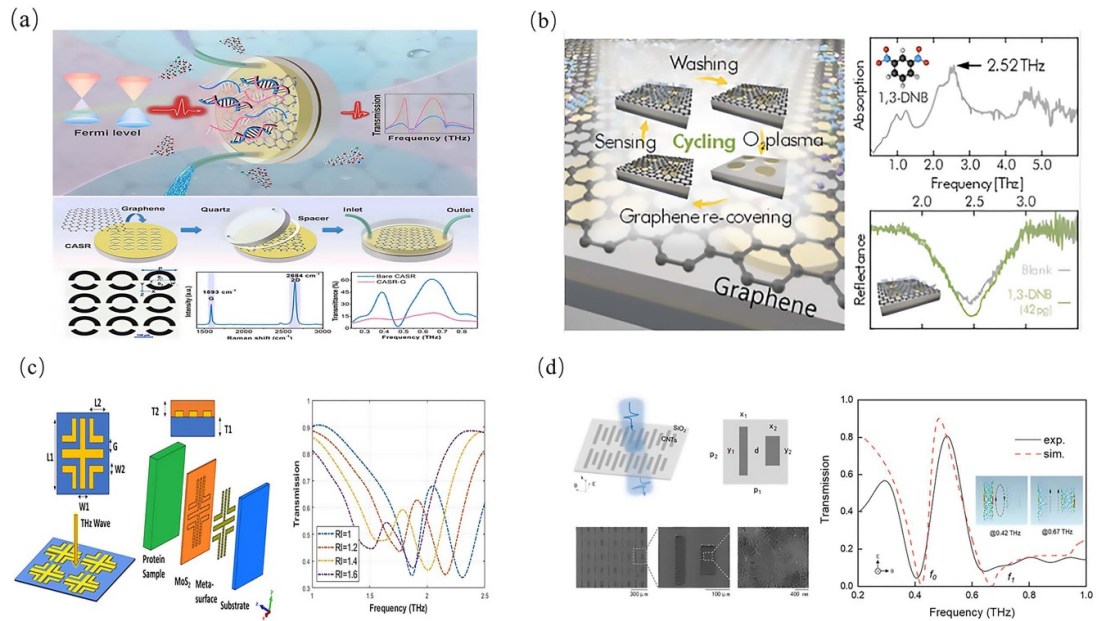


Figure 8. Terahertz metasurface sensor combining with two-dimensional materials or carbon nanotubes. (a) Sensitive DNA detection using graphene-assisted terahertz microfluidic metasensor. Reproduced from [96]. © IOP Publishing Ltd. All rights reserved. (b) Graphene-Enhanced refreshable metasurface expands analytes of terahertz label-free sensing and achieves picogram limit of detection. Reprinted from [97], Copyright (2021), with permission from Elsevier. (c) Ultrahigh sensitive terahertz metasurface with MoS₂ for refractive index biosensing. © (2024) IEEE. Reprinted, with permission, from [99]. (d) Chiral detection based on carbon nanotube metasurface sensor. Reprinted from [100], Copyright (2024), with permission from Elsevier.

[101]. Compared to conventional noble metal materials, CNT films exhibit lower absorption loss and higher field enhancement factors, leading to substantially improved sensor sensitivity and lower detection limits. For example, Yin *et al* employed chemically modified single-walled carbon nanotubes (SWNTs) as a metasurface to enhance terahertz sensing performance [100], as shown in figure 8(d). The SWNT metasurface also functioned as a chiral stationary phase, selectively binding one of two enantiomers. This configuration enables chiral molecule identification by monitoring the optical response of the chemically modified SWNT metasurface. Specifically, the measured spectra display highly selective features in spectral regions corresponding to different chiral responses, a result of the binding affinity differences between the functionalized SWNTs and the chiral molecules.

3.6. Comparison of terahertz metasurface sensors with different materials

Different functional materials exhibit distinct advantages for terahertz sensing. Metal nanoparticle-doped metasurface sensors have achieved a detection limit of 100 pg ml⁻¹ for imidacloprid solutions [70], while graphene-integrated metasurface sensors further reduce the detection limit to 42 pg ml⁻¹ for 1,3-DNB solutions [98]. CNT-based metasurface sensors primarily enable chiral molecule recognition [100], and the detailed comparison of sensors incorporating other materials are summarized in table 2. Although metallic metasurfaces support strong resonances, intrinsic ohmic losses limit *Q*-factor enhancement, and nanoparticle doping increases fabrication complexity and cost. In contrast, graphene, owing to its high carrier mobility and tunable conductivity, significantly improves sensitivity and dynamic tunability, making it particularly suitable for liquid-phase biosensing. The unique layered structure of molybdenum disulfide endows it with controllable band characteristics, further enhancing its absorption and modulation ability for terahertz waves, providing an ideal material platform for efficient and integrated terahertz sensing devices. CNTs offer high biocompatibility and strong terahertz interactions, enabling selective chiral detection through chemical functionalization, while microfluidic integration effectively suppresses water absorption losses. Overall, graphene and CNTs exhibit greater potential for terahertz sensing, with graphene favoring high-sensitivity and dynamically tunable detection and CNTs excelling in chiral recognition and low-loss sensing.

Table 2. Parameters of terahertz metasurface sensors with different materials.

	Metal [62]	Dielectrics [83]	Molybdenum disulfide [99]
Operating frequency band	0.3–0.7 THz	0.1–2.5 THz	0.1–2.5 THz
Sensitivity	0.079 THz RIU ^{−1}	0.230 THz RIU ^{−1}	0.661 THz RIU ^{−1}
Q-factor		412	8.45

4. Application of terahertz metasurface sensors

4.1. Detection of proteins and carbohydrates

Proteins are essential building blocks of human cells and tissues, while carbohydrates serve as the body's primary energy source. Their synergistic interplay supports metabolism, immune defense, and overall physiological function. Given their critical roles, accurate detection of proteins and carbohydrates is of great importance [102–104]. Terahertz metasurface sensors provide an efficient and non-invasive approach to detect these biomolecules by monitoring changes in the dielectric environment, without requiring complex sample preparation [105]. This capability positions them as a transformative technology for biomedical diagnostics, disease biomarker screening, and personalized health monitoring [106, 107].

For instance, Cui *et al* introduced a novel approach based on an all-silicon terahertz metasurface [108], as depicted in figure 9(a). The structure was experimentally validated as a highly sensitive biosensor for detecting Cry1Ac toxin produced by *Bacillus thuringiensis*. Regression analysis revealed strong correlations between the toxin protein concentration and the metasurface resonance signals, with coefficients of 0.8988 for resonance intensity and 0.9238 for resonance frequency. The influence of the analyte was systematically investigated by examining variations in both layer thickness and dielectric constant on the resonance amplitude and frequency shift. Furthermore, the reliability and stability of the metasurface were confirmed under controlled time, temperature, and humidity conditions. In another study, Liu *et al* designed a polarization-independent terahertz ring-shaped sensor by integrating four Taiji resonators into a cyclic unit [109], as illustrated in figure 9(b). The sensor demonstrates high sensitivity to ultrathin analyte layers and refractive index changes, achieving a sensitivity of 0.258 THz RIU^{−1} for a 4 μm analyte thickness in the 1.345 THz frequency range. The sensor was successfully employed to detect low concentrations (0.0001–10 mg ml^{−1}) of A β protein, a biomarker associated with Alzheimer's disease.

For the detection of sugar molecules, Ishak *et al* developed a metasurface sensor featuring a circular groove structure [110], as shown in figure 9(c). This sensor is capable of detecting concentrations of sugar, glucose, and sucrose as low as 0.2%. Experimental results indicate that the sensing performance for glucose and sucrose is improved by approximately 25% and 13%, respectively, at resonance frequencies of 1.98 THz and 1.80 THz. In a complementary approach, Hasebe *et al* demonstrated label-free terahertz sensing of *in vivo* molecular binding using a metasurface composed of a thin metal mesh and a polyvinylidene fluoride membrane [111], as depicted in figure 9(d). Their study revealed that both the frequency shift and transmittance attenuation depend on the binding amount in lectin–sugar interactions, enabling the detection of biotin at quantities as low as 0.17 pg mm^{−2}.

4.2. Detection of DNA

As the blueprint of genetic information, DNA precisely stores and transmits all biological instructions within human cells [112–114]. Given its critical role, accurate DNA detection is essential [115, 116]. Terahertz metasurface sensors exploit the sensitivity of terahertz waves to the dielectric properties of biomolecules, capturing characteristic signals arising from conformational changes and intermolecular interactions in nucleic acids. Coupled with the enhanced electromagnetic field control provided by the metasurface, this approach enables label-free, non-invasive, and highly sensitive detection [117–119].

For DNA sensing, Luo *et al* developed a metal–graphene hybrid terahertz metasurface [120], shown in figure 10(a), for the sensitive and enzyme-free detection of circulating tumor DNA (ctDNA) in plasma samples from pancreatic cancer patients. The amplification effect enabled by gold nanoparticles (AuNPs) on the metasurface was investigated both experimentally and theoretically. The AuNPs enhance charge injection into the graphene film, leading to a significant change in the gradient transmittance index of the metasurface under terahertz irradiation. This mechanism was exploited in a detection

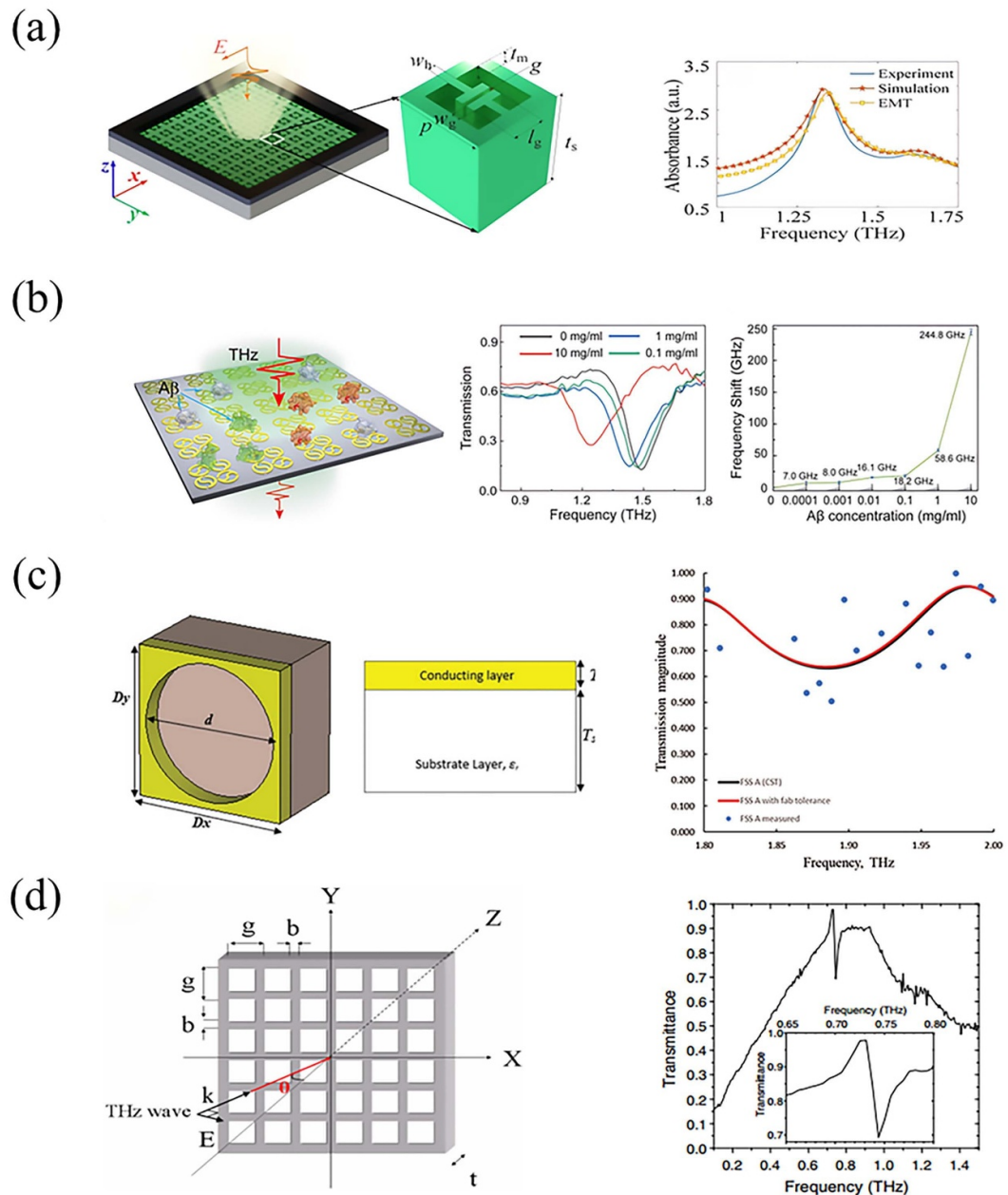


Figure 9. Terahertz metasurface sensor detection of proteins and carbohydrates (a) Sensing performance of an all-silicon terahertz metasurface chip for Cry1Ac protein. Reproduced with permission from [108]. © 2022 Chinese Laser Press. (b) Terahertz Taichi-like micro-ring sensors based on toroidal dipole resonance for concentration detection of Aβ protein. Reproduced from [109]. CC BY 4.0. (c) Detection of low sugar concentration solution using frequency selective surface. Reproduced from [110]. CC BY 4.0. (d) Unlabeled terahertz sensing for detection of sugar by binding of living related molecules. Reprinted from [111], Copyright (2011), with permission from Elsevier.

scheme that triggers a cascade hybridization chain reaction on magnetic beads in the presence of target ctDNA, achieving dual signal amplification—both chemical and optical. In addition to exhibiting high sensitivity down to the subcellular level and single-nucleotide mismatch selectivity, the proposed method effectively distinguishes cancer patients from healthy individuals by identifying and quantifying target ctDNA. In a related study, Lee *et al* utilized a graphene-bound nanogroove structure supporting terahertz resonance to detect ultralow concentrations of ssDNA [121], as illustrated in figure 10(b). The combination of graphene's resonant behavior and its tunable electro-optical properties overcomes the substantial scale mismatch—on the order of 10 000:1—between terahertz wavelengths and the physical dimensions of solid-state DNA molecules. Ultrahigh sensitivity is achieved through a marked increase in the absorption cross-section of graphene sheets and target biomolecules, resulting from strongly enhanced terahertz

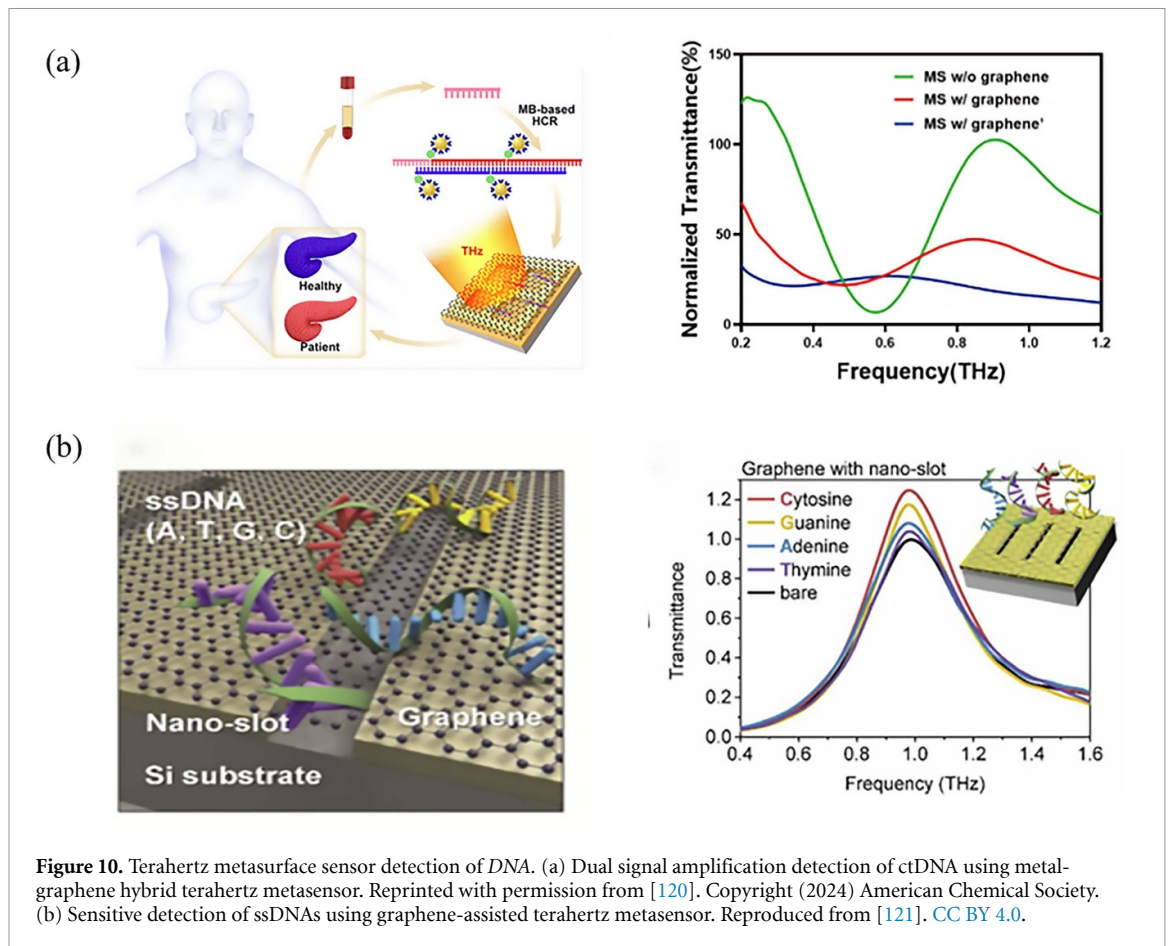


Figure 10. Terahertz metasurface sensor detection of DNA. (a) Dual signal amplification detection of ctDNA using metal-graphene hybrid terahertz metasensor. Reprinted with permission from [120]. Copyright (2024) American Chemical Society. (b) Sensitive detection of ssDNAs using graphene-assisted terahertz metasensor. Reproduced from [121]. CC BY 4.0.

fields inside the nanogrooves at resonance. As a result, clearly distinguishable terahertz optical signals are observed for different types of ssDNA.

4.3. Detection of bacteria and viruses

Although bacteria and viruses are often regarded as pathogens, they also play indispensable roles in human health. Studies have shown that certain bacteria and viruses can activate immune responses and strengthen the host's defenses against other pathogens. Together, they form a dynamic equilibrium within the human microbiome [122–124]. Given their significant impact, accurate detection of bacteria and viruses is of great importance [125, 126].

Terahertz metasurface sensors, leveraging their unique capabilities in electromagnetic field manipulation, high sensitivity, fingerprint spectral characteristics, and broadband operation, enable synchronous detection and dynamic monitoring of multiple pathogens. This offers a revolutionary approach to bacterial and viral detection, providing an efficient and reliable technical platform for clinical diagnostics, food safety, and environmental monitoring [127].

For bacterial detection, Yoon *et al* introduced a terahertz metasurface sensor [46] capable of classifying and identifying diverse microorganisms—including mold, yeast, and bacteria—as illustrated in figure 11(a). The refractive index of fungi ranges between 1.24 and 1.85, while that of bacteria is higher, lying between 2.75 and 4.11, a range relatively close to that of water. Yeast exhibits an even higher dielectric constant, between 5.63 and 5.97, surpassing that of water. These measurements align with low-density microfluidic metasurface results obtained in aqueous environments. Notably, metasurface resonance exhibits a blue shift for mold and bacteria, with mold showing higher sensitivity than bacteria. In contrast, yeast deposition induces a red shift due to its higher dielectric constant compared to water. This approach may represent a significant advance in novel classification tools for microorganisms and in the fabrication of high-sensitivity biosensors, enabling rapid, label-free pathogen detection across various environments. In a separate study, Ma *et al* developed a terahertz metasurface sensor based on electric dipole–toroidal dipole coupling [128], shown in figure 11(b), to achieve fast, non-destructive, label-free identification and highly sensitive quantitative detection of two common pathogenic bacteria. The enhanced toroidal dipole response significantly strengthens light–matter interactions near the microstructure surface, endowing the metasurface sensor with a sensitivity of $0.378 \text{ THz RIU}^{-1}$ and a

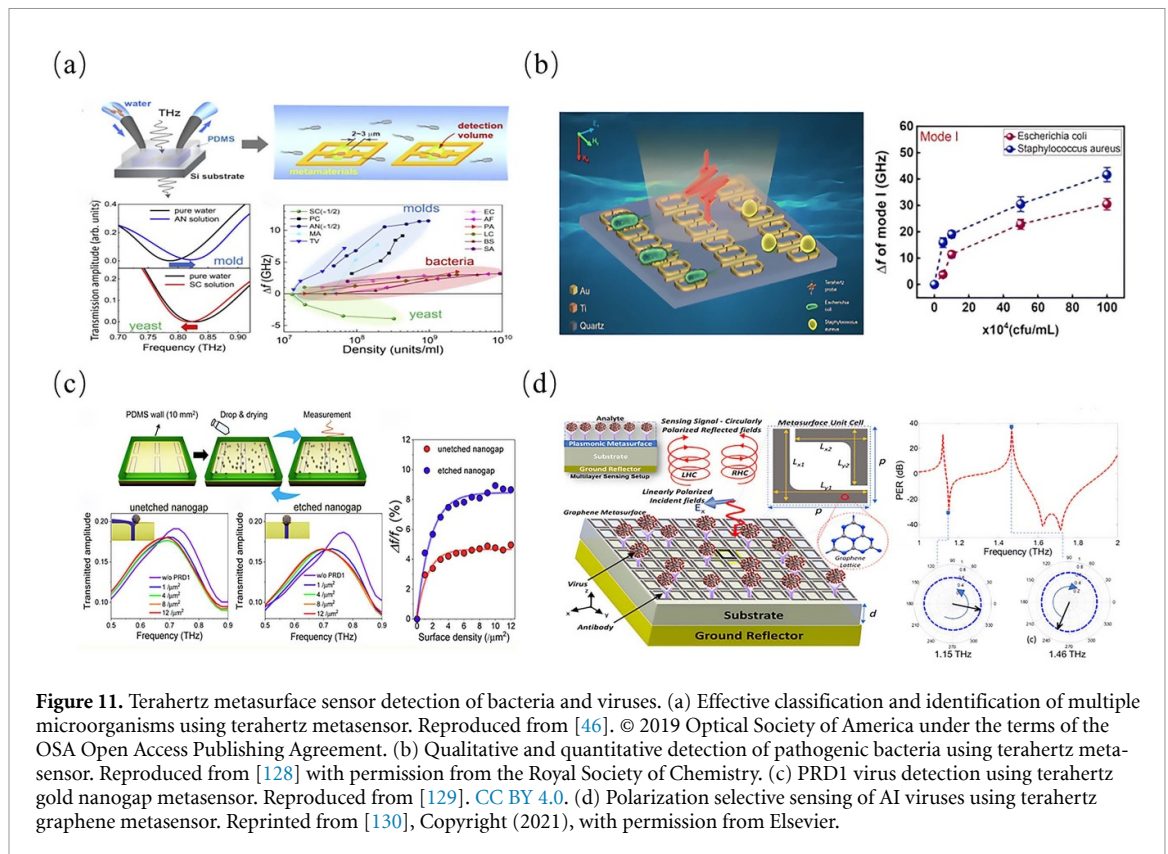


Figure 11. Terahertz metasurface sensor detection of bacteria and viruses. (a) Effective classification and identification of multiple microorganisms using terahertz metasensor. Reproduced from [46]. © 2019 Optical Society of America under the terms of the OSA Open Access Publishing Agreement. (b) Qualitative and quantitative detection of pathogenic bacteria using terahertz metasensor. Reproduced from [128] with permission from the Royal Society of Chemistry. (c) PRD1 virus detection using terahertz gold nanogap metasensor. Reproduced from [129]. CC BY 4.0. (d) Polarization selective sensing of AI viruses using terahertz graphene metasensor. Reprinted from [130], Copyright (2021), with permission from Elsevier.

Q-factor of 21.28. Leveraging these advantages, the biosensor successfully distinguished *Escherichia coli* from *Staphylococcus aureus* and quantitatively detected both across four concentration levels, with a lowest experimental detection limit of 10^4 CFU ml⁻¹ (Colony formation unit per milliliter, core unit for quantitative viable bacterial count in microbiological testing). This work expands the scope of terahertz nanosensors based on interference mechanisms and provides inspiration for further innovations in bio-sensing applications.

For virus detection, Ji *et al* demonstrated ultrasensitive detection of bacteriophage PRD1 using a terahertz metasensor [129], as shown in figure 11(c). The sensor incorporates a virus-scale gold nanogap filled with Al₂O₃, which achieves a 580-fold field enhancement and significantly enhances light-matter interactions. Experimental results showed that the etched nanogap structure produces a twofold increase in resonance shift and a fourfold improvement in sensitivity compared to an unetched nanogap. In a complementary approach, Amin *et al* introduced a comprehensive detection technique based on polarization-state sensing [130], capable of discriminating viral elements with highly similar optical properties. The sensing platform consists of a graphene-based plasmonic metasurface with chiral unit cells, illustrated in figure 11(d). This sensor successfully distinguishes three structurally similar influenza viruses—H1N1, H5N2, and H9N2—by detecting changes in their reflected polarization states.

4.4. Detection of pesticide residues

Pesticides play a vital role in safeguarding food supply, enhancing agricultural product quality, lowering production costs, and curbing the spread of crop diseases. By effectively controlling insects, pathogens, and weeds, they significantly reduce yield losses. Pesticides also help maintain crop health, minimize post-harvest deterioration and the formation of toxic compounds, decrease post-harvest losses, and reduce labor demands for manual weeding and pest management [131]. Given their extensive use, the detection of pesticide residues is of considerable importance. Terahertz metasurface sensors offer an efficient, accurate, and non-destructive solution for pesticide monitoring. Leveraging their unique electromagnetic field control and high sensitivity, they provide powerful technical support for food safety assurance.

For pesticide detection, Xu *et al* demonstrated a metasurface coated with a single layer of graphene [132], as shown in figure 12(a), which operates in the terahertz band and detects pesticides in mixtures by leveraging strong localized interactions between graphene and external molecules. The introduction of single-layer graphene induces a 50% shift in the metasurface's resonance peak. In another study, Hu

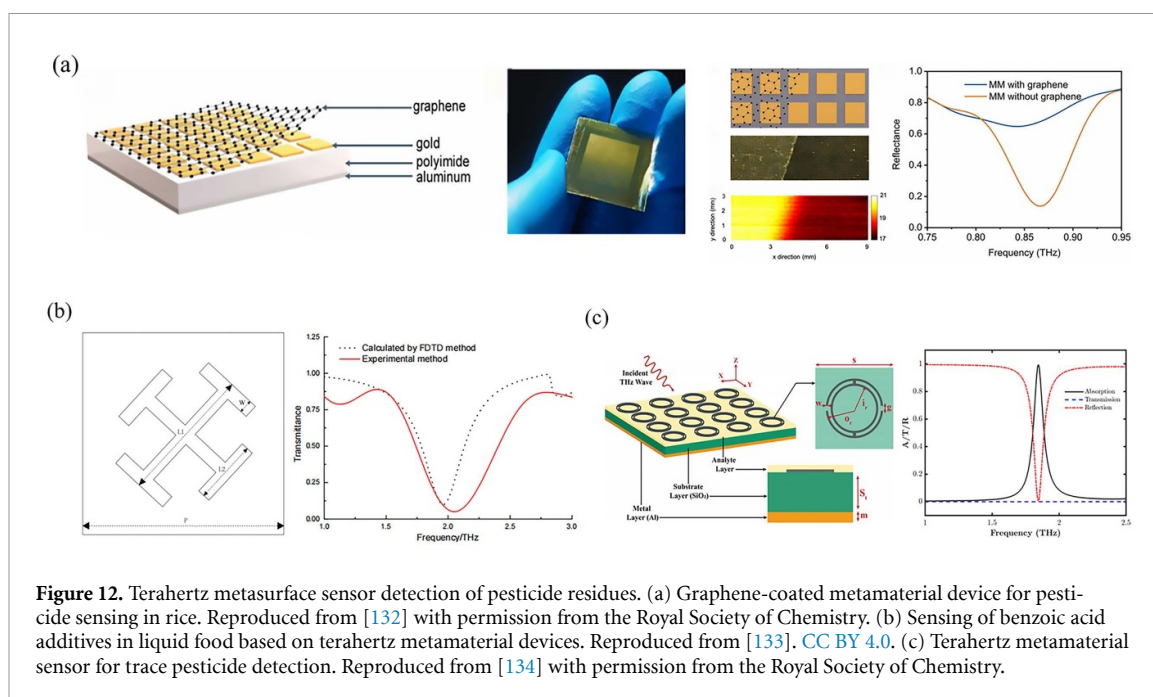


Figure 12. Terahertz metasurface sensor detection of pesticide residues. (a) Graphene-coated metamaterial device for pesticide sensing in rice. Reproduced from [132] with permission from the Royal Society of Chemistry. (b) Sensing of benzoic acid additives in liquid food based on terahertz metamaterial devices. Reproduced from [133]. CC BY 4.0. (c) Terahertz metamaterial sensor for trace pesticide detection. Reproduced from [134] with permission from the Royal Society of Chemistry.

et al designed a terahertz enhanced metasurface to improve the detection of benzoic acid additives using terahertz technology [133], depicted in figure 12(b).

Additionally, Ahmed *et al* developed a unique terahertz metasurface sensor [134] composed of a silicon dioxide substrate, an array of silver structures as a metasurface layer, and an aluminum backplate, illustrated in. This structure exhibits a sensitivity of $0.208 \text{ THz RIU}^{-1}$, a Q -factor of 22.46, and is capable of detecting four different pesticides—IMD, CRE, DETT, and DMDT—in wheat samples at concentrations as low as 0.1 ng. As the pesticide content in the wheat increases, the resonance frequency exhibits a redshift and the absorption peak decreases proportionally. Together, these studies demonstrate significant potential for applications in food safety, agricultural quality control, environmental pollution monitoring, and public health protection.

4.5. Comparison of terahertz metasurface sensors for different application targets

Design strategies are strongly application-oriented, reflecting the distinct requirements of different target analytes. For protein and carbohydrate detection, sensors typically exploit molecular fingerprint absorption by aligning metasurface resonant modes with characteristic spectral features, enabling high-sensitivity recognition and detection of low concentrations ($0.0001\text{--}10 \text{ mg ml}^{-1}$), such as amyloid- β proteins [109]. DNA detection generally relies on chemical functionalization of graphene or CNTs to improve selectivity through specific molecular binding [120], consistent with the material-driven enhancement strategies discussed above. In contrast, the detection of bacteria and viruses imposes more stringent sensitivity requirements; therefore, high Q -factor metasurfaces combined with surface-immobilized antibodies are commonly employed, achieving selective detection of *E. coli* in the presence of *S. aureus* with a detection limit of 10^4 CFU ml^{-1} [128]. For pesticide detection, robustness against interference and broadband spectral coverage are emphasized, and multi-resonant metasurface designs are widely adopted to distinguish target species in complex matrices. Overall, biomolecular sensing prioritizes a trade-off between sensitivity and selectivity through material functionalization and resonance engineering, whereas environmental pollutant detection focuses on anti-interference capability and wide-band response, complementing the material-dependent performance characteristics of terahertz metasurface sensors.

5. Conclusion and prospects

Currently, terahertz metasurface sensors primarily rely on the refractive index properties, fingerprint spectra (i.e. characteristic absorption peaks), or biological features of analytes to achieve precise identification, effective detection, and sensitive sensing of target substances. This review has summarized various

detection strategies based on terahertz metasurface sensors from multiple perspectives, including refractive index sensors operating on different principles, metasurfaces fabricated from diverse materials, and sensors designed for various application targets.

Despite the considerable potential of terahertz metasurface sensors in biomedical and security detection, their development still faces several challenges as terahertz technology advances. The core issues and corresponding solutions are outlined as follows:

First, the strong absorption of terahertz waves by aqueous solutions and other liquid environments leads to severe signal attenuation. This can be mitigated by integrating microfluidic chips to shorten the optical path or by optimizing the structural design to reduce losses. Further optimization of the sensing architecture is also essential. In addition, the adoption of a wider range and newer types of flexible materials can effectively minimize substrate-induced terahertz wave attenuation, ultimately enhancing the signal-to-noise ratio.

Second, temperature fluctuations readily cause frequency shift interference. To address this, multiple approaches can be pursued. On the hardware side, selecting materials with low thermal expansion coefficients and high temperature stability for key sensor components can minimize the physical impact of thermal variations. On the signal-processing side, introducing temperature compensation circuits that monitor ambient temperature in real time and generate counteracting signals can offset frequency drift. Furthermore, intelligent control algorithms can be implemented to dynamically adjust sensor parameters according to temperature changes, ensuring stable and accurate performance across varying thermal conditions.

Third, nonspecific adsorption of biomolecules compromises detection accuracy. Surface functionalization offers a viable solution—for instance, applying anti-fouling coatings such as polyethylene glycol, which forms a hydration layer to repel nonspecific molecules. Molecular imprinting techniques can also be employed to create cavities on the sensor surface that are complementary in shape and functionality to the target biomolecules, enabling selective binding. Optimizing the detection environment, including adjusting ionic strength and pH, can further reduce nonspecific adsorption and improve detection accuracy.

Fourth, material limitations and high costs pose constraints. Breakthroughs can be sought through multiple dimensions. In material research, efforts should focus on developing novel, low-loss, highly stable, and cost-effective materials, such as specific organic polymers or composites, to replace conventional expensive inorganic ones. Process optimization can also improve material utilization efficiency. In terms of cost reduction, large-scale production can help distribute research and manufacturing expenses. Advanced micro-nano fabrication technologies can enable sensor miniaturization and integration, thereby reducing material usage and production steps. Moreover, establishing industry–academia–research collaborations can facilitate resource and technology sharing, accelerate technology translation, lower overall costs, and promote wider adoption of terahertz sensors.

Fifth, noise can significantly interfere with terahertz metasurface sensors. To mitigate this, traditional metals can be replaced with low-loss materials to reduce thermal noise from electron scattering. Adsorption specificity of target molecules can be enhanced through chemical modification to suppress non-specific background signals. Finally, advanced signal processing techniques—such as filtering, wavelet denoising, and machine learning algorithms—can extract characteristic signals. Once trained, these algorithms can distinguish noise from true signals, improving detection accuracy and reliability.

Looking forward, quantum sensing represents a promising direction. This technology exploits unique quantum properties—such as superposition, entanglement, and coherence—to achieve high-precision measurements of physical quantities [135–137]. By manipulating quantum states of microscopic particles (e.g. spin, energy level, or phase) and observing their evolution in response to target physical quantities, relevant information can be extracted [138–141]. The integration of terahertz metasurface sensors with quantum sensing is expected to enable ultra-high sensitivity detection of magnetic fields, electric fields, temperature, gravity, and other parameters [142, 143]. This synergy may surpass the accuracy limits of conventional sensors in areas such as navigation and positioning, medical imaging, environmental monitoring, and fundamental scientific research.

Further research and innovation are essential to explore diverse sensor architectures and realize more efficient and sensitive detection capabilities. Advances in sensor design and technology will help overcome existing limitations and pave the way for developing efficient, multifunctional terahertz sensors capable of identifying and quantifying a broader range of substances in complex environments. With continued technological breakthroughs, terahertz metasurface sensors are poised to become a key enabler of intelligent advancement across multiple fields.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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